Minecraft Tree World Resilience Cup

Design a climate-resilient landscape

At a glance: Design a landscape that survives stress because of how it is built, not luck. Plan about 80 minutes for the core block; 4 teams of 4 students.

LEARNING OBJECTIVES

BY THE END OF THIS ACTIVITY, EACH STUDENT CAN:

- **Explain the core science of this challenge: ecosystems, biodiversity, climate resilience, systems design (Understand).**
- **Design a fair test by controlling one variable while holding others constant (Apply).**
- **Use their own data to decide whether a redesign improved the result (Analyze).**
- **Defend a final claim with specific evidence (Evaluate).**

MATERIALS FOR 16 STUDENTS AND 4 TEAMS

- **Minecraft Education devices: one per team (4, plus 1 spare), each with the resilience world template loaded; license is per user, not per device**
- **paper-grid fallback kit: one per team (4, plus 1 spare) for device or login failure**
- **rubric cards: one per team (4, plus 1 spare)**
- **display timer: 1, plus the projector**

PREP BEFORE STUDENTS ARRIVE (15 TO 20 MIN)

01 Set up one station per team with the materials above, sorted and ready.

02 Stage the scoring sheet and any reference values before testing begins.

3 No PPE is needed: this is a dry, seated, screen-based design activity with no soil, water, or chemicals. Instead, confirm each team's device is signed in with the world template loaded, and stage one paper-grid fallback kit per team in case a device or login fails.

01 Load the activity slides and ready the projected timer.

MINUTE-BY-MINUTE FACILITATION

Phase 1 Hook

0:00 - 0:05

Pose the mission as a challenge: "Design a landscape that survives stress because of how it is built, not luck." Take a quick prediction by show of hands.

Phase 2 Prediction

0:05 - 0:12

Before any materials move, each team writes one prediction and one reason on the handout. No building yet.

Phase 3 Constraints

0:12 - 0:18

Set the rules: pick one stress to design against, justify every feature with a real nature-based strategy, label each strategy on the build, and follow the device-use notes.

Phase 4 Build or solve

0:18 - 0:44

Teams work through the steps on the student handout. Circulate, ask what they are testing, and resist solving it for them.

Phase 5 Build the resilient landscape

Teams pick one stress (flood, drought, or heat) and build diversity, connected green space, shade, and water control, labeling each feature with the real strategy it represents.

Phase 6 Refine one weakness

Each team identifies the part of its landscape that would fail first under its chosen stress, makes one targeted change to fix it, and notes the before-and-after.

Phase 7 Score, debrief, cleanup

last 12 min

Teams submit a score slip, answer a debrief question with evidence, sign out of the device, and reset it (or collect the paper-grid sheets) for the next team. No PPE to remove; wipe shared device surfaces only if site policy requires.

TROUBLESHOOTING

Problem: A team changed several things at once. **Fix:** Have them reset to one variable before the next reading counts; treat it as a data-quality coaching moment.

Problem: A device or login fails. Fix: Switch the team to the spare device or the paper-grid fallback kit immediately so they keep designing; do not let a tech failure cost build time.

Problem: Readings vary between trials. **Fix:** Standardize the procedure: same conditions, same technique, average several trials.

Problem: A team finishes early. **Fix:** Push the extension prompt below and ask them to quantify their improvement.

DIFFERENTIATION: THREE-TIER SUPPORT PROMPTS

Tier 1 (stuck at the start):

- "What is the one science idea behind this challenge?" Point them to resilience by design.

Tier 2 (building but not reasoning):

- "What single change could improve your result without changing anything else?"

Tier 3 (ready to extend):

- "Run a second version that isolates a different variable and compare the two with data."

DEBRIEF QUESTIONS (PICK 2 TO 3)

- Which feature would help most in a real storm?
- Why does diversity add resilience?
- What in your build would fail first, and how would you fix it?

SCORING RUBRIC (100 POINTS)

SCORING CRITERION	POINTS
Resilience features	30
Biodiversity support	25
Realism	20
Explanation	15
Build polish	10
Total	**100**

CLEANUP AND SOURCES

Return reusable items to the bin, dispose of consumables, wipe surfaces, and collect score slips.

SOURCES Minecraft Education, Biodiversity resources (https://education.minecraft.net/en-us/resources/science/biodiversity); licensing is per user (https://education.minecraft.net/en-us/licensing).